

DETAILED SYLLABUS

Theory of Computation L-T-P: 3-0-0 Credit-3 MAxxxxx

Prerequisites: Design and Analysis of Algorithms

Category: Existing and revised

Syllabus:

1. **Finite automata and regular languages:** Finite automata, regular languages and expressions, closure properties of regular languages, subset construction, product construction, state minimization, pumping lemma for finite automata, application of the pumping lemma, Myhill-Nerode theorem, two-way finite automata **(9 lectures)**
2. **CFGs and CFLs:** Context-free grammars and languages, derivation tree/parse tree, Chomsky Normal Form, Greibach Normal Form, push down automata, Pumping lemma for CFLs, closure properties of CFLs, Cocke-Kasami-Younger algorithm **(9 lectures)**
3. **Turing machines:** Deterministic and non-deterministic Turing machines, decidability and undecidability, reductions, Rice's theorem, lambda calculus, Godel's incompleteness theorem **(9 lectures)**
4. **Computational Complexity:** Linear speedup theorem, complexity classes, NP-completeness, Cook-Levin theorem, NP-hard problems on graphs, Savitch's theorem **(9 lectures)**

References:

- 1) Automata and Computability: Dexter C. Kozen
- 2) Theory of Computation: Dexter C. Kozen
- 3) Computational Complexity a Modern Approach - Sanjeev Arora, Boaz Barak
- 4) Discrete Mathematics and its Applications - Kenneth H. Rosen

Teachers: Prof. Ratna Dutta, Dr. Bodhayan Roy
